Math 110 — Week 8

Solutions

HW8 P1.

Idea. Differentiation lowers the degree, so enough derivatives send every polynomial to 0; the index is the least k with $D^k = 0$, and x^n is the basis vector that needs the most derivatives to reach 0.

1. (a) D lowers the degree by 1 and sends constants to 0, so $D^{n+1}p = 0$ for every $p \in \mathcal{P}_n(\mathbb{C})$, while $D^n x^n = n! \neq 0$. So D is nilpotent of index $n+1$.
2. (b) $E(0, D) = \operatorname{null} D$ is the constants, so the geometric multiplicity is 1. D is nilpotent, so $G(0, D) = \mathcal{P}_n(\mathbb{C})$ and the algebraic multiplicity is $n+1$.
3. (c) Start from x^n and differentiate repeatedly; dividing by factorials makes D send each new basis vector exactly to the previous one. Let $f_k = x^k/k!$ for $k = 0, \ldots, n$. Then $Df_0 = 0$ and $Df_k = x^{k-1}/(k-1)! = f_{k-1}$.
4. In the basis $f_0, f_1, \ldots, f_n$, column k has a single 1 in row $k-1$: the matrix is the single block $J_{n+1}(0)$.

HW8 P2.

Idea. Translate the data into blocks: geometric multiplicity = number of blocks, algebraic multiplicity = total size, largest block = exponent in m.

1. (a) Eigenvalue 2: one block of total size 3, so $J_3(2)$. Eigenvalue -1: two blocks of total size 2, so $J_1(-1) \oplus J_1(-1)$:

$$\begin{pmatrix} 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{pmatrix}, \qquad m = (x-2)^3(x+1).$$

2. (b) Two blocks for 2 with total size 3: $J_2(2) \oplus J_1(2) \oplus J_1(-1) \oplus J_1(-1)$, the matrix in (a) with the $(2,3)$ entry changed to 0. The largest block for 2 has size 2, so $m = (x-2)^2(x+1)$.

HW8 P3.

Idea. $J - I$ is the shift $e_3 \mapsto e_2 \mapsto e_1 \mapsto 0$, so its powers are easy; for J^m, split J into the commuting parts I and $J - I$ and use the binomial theorem, which stops at $(J-I)^2$.

1. (a) $N = J - I$ has $Ne_1 = 0$, $Ne_2 = e_1$, $Ne_3 = e_2$. So $\operatorname{null} N = \operatorname{span}(e_1)$. $N^2 e_3 = e_1$ and $N^2 e_1 = N^2 e_2 = 0$, so $\operatorname{null} N^2 = \operatorname{span}(e_1, e_2)$. $N^3 = 0$, so $\operatorname{null} N^3 = \mathbb{F}^3$.
2. (b) $G(1, J) = \operatorname{null}(J-I)^3 = \mathbb{F}^3$. The geometric multiplicity is $\dim \operatorname{null}(J-I) = 1$; the algebraic multiplicity is $\dim G(1, J) = 3$.
3. (c) $J = I + N$ with $IN = NI$ and $N^3 = 0$, so by the binomial theorem

$$J^m = I + mN + \binom{m}{2} N^2 = \begin{pmatrix} 1 & m & \binom{m}{2} \\ 0 & 1 & m \\ 0 & 0 & 1 \end{pmatrix},$$

with $\binom{m}{2} = \frac{m(m-1)}{2}$; for $m = 1$ this is J.

HW8 P4.

Idea. Null spaces of a block diagonal matrix add block by block, and for one block, $J_j(0)^k$ is a shift by k places. Differences of the d_k count blocks of size $\geq k$; the total dimension determines the largest size.

1. (a) For one block $J_j(0)$, $\dim \operatorname{null} J_j(0)^k = \min(k, j)$. Adding over the blocks of sizes 3 and 2: $k = 0$: 0; $k = 1$: $1 + 1 = 2$; $k = 2$: $2 + 2 = 4$; $k = 3$: $3 + 2 = 5$; $k = 4$: $3 + 2 = 5$.
2. (b) The number of blocks of size $\geq k$ is $\dim \operatorname{null} N^k - \dim \operatorname{null} N^{k-1}$ (Prop. 77): $2 - 0 = 2$ of size ≥ 1, $4 - 2 = 2$ of size ≥ 2, $5 - 4 = 1$ of size ≥ 3. So there are two blocks, both of size at least 2, exactly one of size at least 3.
3. The three numbers alone do not fix the larger size ($J_4(0) \oplus J_2(0)$ on $\mathbb{C}^6$ also gives $2, 4, 5$). But N acts on the 5-dimensional space of (a): the sizes are 2 and $s \geq 3$ with $2 + s = 5$, so $s = 3$.

HW8 P5.

Idea. For each eigenvalue, make the table of d_k and take differences (Prop. 77).

1. Eigenvalue 1: $d_1, d_2, d_3 = 1, 2, 2$, so 1 block of size ≥ 1, 1 of size ≥ 2, none of size ≥ 3: $J_2(1)$.
2. Eigenvalue 2: $d_1, \ldots, d_4 = 2, 3, 4, 4$, so 2 blocks of size ≥ 1, 1 of size ≥ 2, 1 of size ≥ 3, none of size ≥ 4: $J_3(2)$ and $J_1(2)$.
3. Check: $2 + 3 + 1 = 6$. So T has the Jordan form $J_2(1) \oplus J_3(2) \oplus J_1(2)$.

HW8 P6.

Idea. Each quantity is read off the blocks: total size, largest size, number of blocks.

1. (a) The blocks for 4 have total size $3 + 2 = 5$ and the block for -1 has size 1, so $\chi = (x - 4)^5(x + 1)$.
2. (b) The largest blocks are 3 (for 4) and 1 (for -1), so $m = (x - 4)^3(x + 1)$.
3. (c) There are two blocks for 4, so $\dim E(4) = 2$, while $\dim G(4) = 5$; for -1 both are 1.
4. (d) m has the repeated root 4, so the operator is not diagonalizable (Week 7, Prop. 68).

HW8 P7.

Idea. Everything about a block diagonal matrix splits into blocks, so count block by block; for one block, $J_j(0)^k$ is a shift by k places.

1. Choose a basis in which N is in Jordan form with blocks $J_{j_1}(0), \ldots, J_{j_r}(0)$. N^k is block diagonal with blocks $J_{j_i}(0)^k$, so $\dim \operatorname{null} N^k = \sum_i \dim \operatorname{null} J_{j_i}(0)^k$.
2. On one block with basis $e_1, \ldots, e_j$, $J_j(0)^k e_l = e_{l-k}$ for $l > k$ and 0 for $l \leq k$. So $\operatorname{null} J_j(0)^k = \operatorname{span}(e_1, \ldots, e_{\min(k,j)})$, of dimension $\min(k, j)$.
3. $\min(k, j) - \min(k - 1, j)$ is 1 if $j \geq k$ and 0 if $j < k$. Summing over the blocks, $\dim \operatorname{null} N^k - \dim \operatorname{null} N^{k-1}$ counts the blocks with $j_i \geq k$.

G1.

Idea. Get the eigenvalues from the blocks, then make the d_k table, then build a chain from a top outside $\operatorname{null}(A-2I)$.

1. The upper-left 2×2 block has trace 4 and determinant 4, so its characteristic polynomial is $(x-2)^2$, and the lower block is (2). A is block diagonal, so $G(\lambda, A)$ is the direct sum of the generalized eigenspaces of the blocks (T9), and the algebraic multiplicities add: $\chi_A = (x-2)^3$, and 2 is the only eigenvalue.
2. The matrix
$$A - 2I = \begin{pmatrix} -1 & 1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$
has rank 1, so $E(2, A) = \{y = x\} = \operatorname{span}\big((1,1,0),(0,0,1)\big)$ and $d_1 = 2$.
3. $(A-2I)^2 = 0$, so $d_2 = 3$ and $G(2, A) = \mathbb{C}^3$. Blocks: 2 of size ≥ 1, 1 of size ≥ 2: $J_2(2) \oplus J_1(2)$, and $m_A = (x-2)^2$.
4. Jordan basis: the top $e_1 \notin \operatorname{null}(A-2I)$ has the bottom $(A-2I)e_1 = (-1,-1,0)$; then add the eigenvector e_3, independent of that bottom. In the basis $(-1,-1,0), e_1, e_3$ the matrix is $J_2(2) \oplus J_1(2)$. Check: $Ae_1 = (1,-1,0) = 2e_1 + (-1,-1,0)$.

G2.

Idea. As for eigenspaces: S commutes with every polynomial in T, so it preserves the null spaces of polynomials in T.

1. $ST = TS$ gives $S(T-\lambda I) = (T-\lambda I)S$, and then $S(T-\lambda I)^n = (T-\lambda I)^n S$.
2. $G(\lambda, T) = \operatorname{null}(T-\lambda I)^n$ (Prop. 70). If $(T-\lambda I)^n v = 0$, then $(T-\lambda I)^n Sv = S(T-\lambda I)^n v = 0$.

G3.

Idea. (a) $(\lambda I + N)v = 0$ says v is an eigenvector of N for $-\lambda$, and 0 is the only eigenvalue of N. (b) With one eigenvalue there is one generalized eigenspace, and it is everything.

1. (a) If $\lambda \neq 0$ and $(\lambda I + N)v = 0$, then $Nv = -\lambda v$; since the only eigenvalue of a nilpotent N is 0 (if $Nv = \mu v$ with $v \neq 0$, then $0 = N^k v = \mu^k v$, so $\mu = 0$), and $-\lambda \neq 0$, $v = 0$. So $\lambda I + N$ is injective, hence invertible (T2).
2. If $\lambda = 0$, then N is not injective: $V \neq 0$, and $N^k = 0$ means some $N^j v \neq 0$ has $N^{j+1}v = 0$.
3. For $\lambda \neq 0$ the inverse is explicit:
$$(\lambda I + N)^{-1} = \lambda^{-1} \sum_{j=0}^{n-1} (-N/\lambda)^j.$$
4. (b) By Prop. 73, $V = G(\lambda, T) = \operatorname{null}(T-\lambda I)^n$, so $(T-\lambda I)^n = 0$.

G4.

Idea. Block sizes add to the algebraic multiplicity 4; the largest is the exponent 2 in m_T.

1. The sizes have largest 2 and sum 4: $\{2,2\}$ or $\{2,1,1\}$. So the Jordan form is $J_2(2) \oplus J_2(2)$ or $J_2(2) \oplus J_1(2) \oplus J_1(2)$.
2. They have the same χ_T and m_T; $\dim E(2, T)$ (2 versus 3) tells them apart.

G5.

Idea. Over $\mathbb{C}$, diagonalizable means $E(\lambda, T) = G(\lambda, T)$ for every λ.

1. If $0 \neq v \in G(\lambda, T)$, the hypothesis gives $Tv = \mu v$ for some μ; then $0 = (T-\lambda I)^n v = (\mu - \lambda)^n v$, so $\mu = \lambda$. Hence $G(\lambda, T) \subseteq E(\lambda, T)$; and $E(\lambda, T) \subseteq G(\lambda, T)$ always. So $G(\lambda, T) = E(\lambda, T)$.
2. By Prop. 73, $V = \bigoplus G(\lambda_i, T) = \bigoplus E(\lambda_i, T)$, and bases of the eigenspaces put together form a basis of V (T12 of the handout), which consists of eigenvectors. So T is diagonalizable.

R1.

Idea. Triangular, so χ comes from the diagonal (LADR 8.31); the only candidate for m below χ is $(x-3)(x-1)$.

1. The diagonal is $3, 3, 1$, so $\chi = (x-3)^2(x-1)$ (LADR 8.31) and the eigenvalues are 3 and 1.
2. The candidate of degree 2 fails:
$$(A-3I)(A-I) = \begin{pmatrix} 0 & 2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \neq 0,$$
By Cayley–Hamilton $m \mid \chi$, and m has both roots $3, 1$ (Prop. 66); the divisors of χ with both roots are $(x-3)(x-1)$, which fails, and χ. So $m = \chi = (x-3)^2(x-1)$.
3. The eigenspaces are
$$E(3) = \operatorname{null} \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & -1 \\ 0 & 0 & -2 \end{pmatrix} = \operatorname{span}(e_1), \qquad E(1) = \operatorname{span}\big((-3, 2, 4)\big).$$
Check the second:
$$A \begin{pmatrix} -3 \\ 2 \\ 4 \end{pmatrix} = \begin{pmatrix} -9+2+4 \\ 6-4 \\ 4 \end{pmatrix} = \begin{pmatrix} -3 \\ 2 \\ 4 \end{pmatrix}.$$
4. m has the repeated root 3, so A is not diagonalizable.

R2.

Idea. Read both numbers off the Jordan form: $\dim E(\lambda)$ is the number of blocks, the exponent is the largest block.

1. False. For $J_2(0)$, $\dim E(0) = 1$ but $m = x^2$ has exponent 2.
2. For $J_1(0) \oplus J_1(0)$, $\dim E(0) = 2$ but $m = x$ has exponent 1.